

WEEK 5 – POWER BI EDA REPORT

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Tool Used: Microsoft Power BI

Project: Healthcare Data Transformation & Data Modeling

1. INTRODUCTION

Power BI is a business intelligence and data visualization tool used for transforming, organizing, and analyzing data.

In this Week 5 task, I worked with the **healthcare_dataset.csv** dataset using Microsoft Power BI.

I used **Power Query** to clean and transform the healthcare data and created an **Age Group** column using a Conditional Column. After transforming the data, I organized the data into separate tables and created a structured data model.

I also established **Many-to-One (*:1) relationships** between the related tables.

This task helped me gain practical knowledge of Power Query, data transformation, Conditional Columns, Data Modeling, and Relationships in Power BI.

2. OBJECTIVE

The main objective of this Week 5 task is to prepare and organize the healthcare dataset using Microsoft Power BI.

Objectives:

- To clean and transform healthcare data using Power Query.
 - To apply different data transformation techniques.
 - To create an **Age Group** column using Conditional Column.
 - To organize the healthcare data into meaningful tables.
 - To establish **Many-to-One (*:1) relationships**.
 - To prepare a structured data model for further analysis.
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3. DATASET

For this task, I used the **healthcare_dataset.csv** dataset in Microsoft Power BI.

The dataset contains healthcare-related information such as:

- Patient ID
- Name
- Age
- Gender
- Blood Type
- Medical Condition
- Date of Admission
- Doctor
- Hospital
- Insurance Provider
- Billing Amount
- Room Number
- Admission Type
- Discharge Date
- Medication
- Test Results
- Admission ID

This dataset was used for data transformation and data modeling in Power BI.

Table.AddColumn(#"Cleaned Text1", "Age group", each if [Age] <= 12 then "Child" else if [Age] <= 19 then "Teenager" else if [Age] <= 39						
Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
1 Bobby Jackson	30	MALE	B-	Cancer	31-01-2024	Matthew Smit
2 Leslie Terry	62	MALE	A+	Obesity	20-08-2019	Samantha Da
3 Danny Smith	76	FEMALE	A-	Obesity	22-09-2022	Tiffany Mitch
4 Andrew Watts	28	FEMALE	O+	Diabetes	18-11-2020	Kevin Wells
5 Adrienne Bell	43	FEMALE	AB+	Cancer	19-09-2022	Kathleen Han
6 Emily Johnson	36	MALE	A+	Asthma	20-12-2023	Taylor Newto
7 Edward Edwards	21	FEMALE	AB-	Diabetes	03-11-2020	Kelly Olson
8 Christina Martinez	20	FEMALE	A+	Cancer	28-12-2021	Suzanne Thor
9 Jasmine Aguilar	82	MALE	AB+	Asthma	01-07-2020	Daniel Fergus
10 Christopher Berg	58	FEMALE	AB-	Cancer	23-05-2021	Heather Day
11 Michelle Daniels	72	MALE	O+	Cancer	19-04-2020	John Duncan
12 Aaron Martinez	38	FEMALE	A-	Hypertension	13-08-2023	Douglas Mayr
13 Connor Hansen	75	FEMALE	A+	Diabetes	12-12-2019	Kenneth Fletc
14 Robert Bauer	68	FEMALE	AB+	Asthma	22-05-2020	Theresa Freei
15 Brooke Brady	44	FEMALE	AB+	Cancer	08-10-2021	Roberta Stew
16 Ms. Natalie Gamble	46	FEMALE	AB-	Obesity	01-01-2023	Maria Dough
17 Haley Perkins	63	FEMALE	A+	Arthritis	23-06-2020	Erica Spencer
18 Mrs. Jamie Campbell	38	MALE	AB-	Obesity	08-03-2020	Justin Kim
19 Luke Burgess	34	FEMALE	A-	Hypertension	04-03-2021	Justin Moore
20 Daniel Schmidt	63	MALE	B+	Asthma	15-11-2022	Denise Gallov
21 Timothy Burns	67	FEMALE	A-	Asthma	28-06-2023	Krista Smith
22 Christopher Bright	48	MALE	B+	Asthma	21-01-2020	Gregory Smit
23 Kathryn Stewart	58	FEMALE	O+	Arthritis	12-05-2022	Vanessa New

4. POWER QUERY DATA TRANSFORMATION & AGE GROUP

I used **Power Query Editor** to clean and transform the healthcare dataset before creating the data model.

4.1 Data Transformation Steps

1. Promoted Headers

I promoted the first row as column headers so that each field could be identified correctly.

2. Changed Type

I changed the data types of the columns according to their values, such as text, number, and date.

3. Capitalized Each Word

I formatted the required text values by capitalizing each word to maintain consistent formatting.

4. Removed Errors

I removed error values from the dataset to improve data quality.

5. Trimmed Text

I removed unnecessary spaces from text values using the Trim transformation.

6. Cleaned Text

I cleaned unwanted or non-printable characters from the text data.

7. Uppercased Text

I converted the required text values into uppercase for consistent formatting.

8. Added Index

I added an Index Column to provide sequential index values and support data organization.

9. Removed Columns

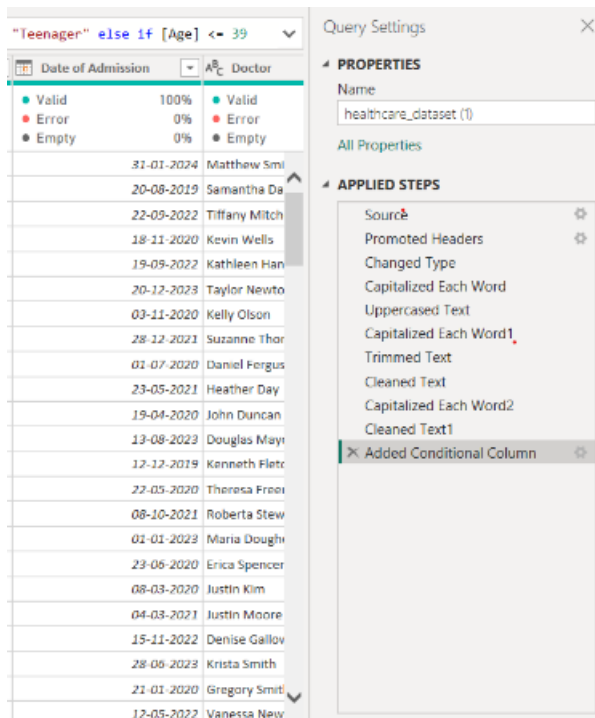
I removed unnecessary columns that were not required for the task.

10. Renamed Columns

I renamed the required columns to make the field names clear and meaningful.

11. Removed Duplicates

I removed duplicate records to improve the consistency and quality of the dataset.



4.2 AGE GROUP USING CONDITIONAL COLUMN

I created a new column named **Age Group** using the **Conditional Column** feature in Power Query.

The Age column was used to classify patients into different age groups.

Age Condition	Age Group
Age $\leq$ 12	Child
Age $\leq$ 19	Teenager
Age $\leq$ 39	Younger Adult
Age $\leq$ 59	Elder Adult
Age $>$ 59	Senior Adult

This classification makes it easier to group patients based on their age.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Age group

	Column Name	Operator	Value ①		Output ①	
If	Age	is less than or equ...	ABC 123 12	Then	ABC 123 Child	...
Else If	Age	is less than or equ...	ABC 123 19	Then	ABC 123 Teenager	
Else If	Age	is less than or equ...	ABC 123 39	Then	ABC 123 Younger Adult	
Else If	Age	is less than or equ...	ABC 123 59	Then	ABC 123 Elder Adult	

Add Clause

Else ①

ABC 123

Senior Adult

OK

Cancel

5. DATA MODELING

After completing the data transformation, I created a structured data model in Power BI.

I organized the healthcare data into the following tables:

Dim_Patient

This table contains patient-related information:

- Patient ID
- Name
- Age
- Age Group
- Gender
- Blood Type

Dim_Admission

This table contains admission and medical-related information:

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number

- Test Results

Billing

This table contains billing and admission-related information:

- Patient ID
- Admission ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider

Creating separate tables helped organize the healthcare data into a structured model for further analysis.

Queries [4]

healthcare_dataset (1)

- Dim_Patient
- Dim_Admission
- Billing

Table: healthcare_dataset (1)

	t	t ₃ Room Number	A _B C Admission Type
	100% 0% 0%	Valid Error Empty	100% 0% 0%
1	18856.28131	328	Urgent
2	33643.32729	265	Emergency
3	27955.09608	205	Emergency
4	37909.78241	450	Elective
5	14238.31781	458	Urgent
6	48145.11095	389	Urgent
7	19580.87234	389	Emergency
8	45820.46272	277	Emergency
9	50119.22279	316	Elective
10	19784.63106	249	Elective
11	12576.79561	394	Urgent
12	7999.58688	288	Urgent
13	43282.28336	134	Emergency
14	33207.70663	309	Urgent

6. RELATIONSHIPS

After creating the tables, I established relationships between the tables using their common key fields.

Relationship 1 – Patient

Billing $\rightarrow$ **Dim Patient**

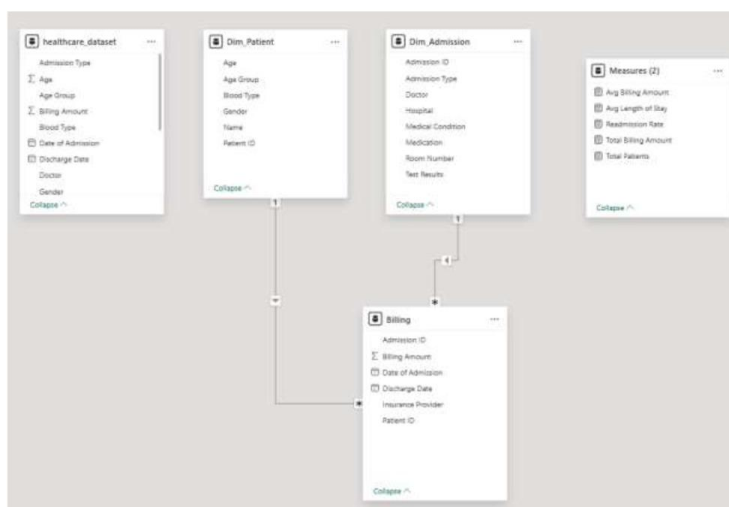
- Common Key: Patient ID
- Cardinality: Many-to-One (*:1)

Relationship 2 – Admission

Billing → Dim_Admission

- Common Key: Admission ID
- Cardinality: Many-to-One (*:1)

These relationships connect the tables and allow related information to be used together during analysis.



7. CONCLUSION

In Week 5, I cleaned and transformed the healthcare dataset using **Power Query** and created an **Age Group** using a Conditional Column.

I then organized the data into **Dim_Patient, Dim_Admission, and Billing** tables and established **Many-to-One (*:1) relationships** between them.

This task helped me gain practical knowledge of **Data Transformation, Data Modeling, Conditional Columns, Power Query, and Relationships in Power BI**.